Socio-Economic Dimensions of the German Energy Transition ("Energiewende")

An Empirical Pipeline Analysis of Energy Poverty in Germany (1995–2025)

Maximilian Starp

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Abstract

The German "Energiewende" represents one of the most significant structural transformations of an industrial economy in modern history. While the expansion of renewable energy is often evaluated through ecological or technological metrics, the socio-economic impact on private households remains a critical area of concern.

This paper presents a empirical analysis of household energy burdens in Germany between 1995 and 2024 using a fully automated data pipeline based on Eurostat datasets. By combining energy price statistics, consumption benchmarks, and a lognormal model of household income distribution, the study estimates the proportion of households experiencing energy poverty under the widely used 10% expenditure threshold.

The results indicate a substantial increase in energy poverty over the observed period and identify a significant contribution of fiscal price components to the total household energy burden. Finally, two political instruments to counteract this development are briefly discussed. After which the economically more sensible one is roughly estimated in fiscal terms.

1 Introduction

Mitigating anthropogenic climate change has become one of the primary objectives of modern economic policy. In Germany, this objective has materialized in the form of the "Energiewende", a long-term transition away from fossil fuels toward renewable energy sources [3]. Since the early 2000s, legislative frameworks such as the Renewable Energy Sources Act (EEG) have accelerated the deployment of renewable electricity generation while restructuring the energy market [5].

While the environmental objectives of the energy transition are widely supported, the socio-economic implications for private households remain contested. Rising electricity and heating costs raise concerns regarding the distributional fairness of climate policy instruments. In particular, consumption-based levies may impose disproportionate burdens on lower-income households.

This paper therefore investigates whether the fiscal components of the German energy transition contribute to rising levels of "energy poverty". Energy poverty describes a condition in which households must allocate a disproportionate share of their disposable income to satisfy basic energy needs such as heating and electricity [13].

The analysis aims to quantify this burden through an integrated empirical data pipeline that combines energy price statistics, household consumption benchmarks, and income distribution modeling. Rather than attempting to establish strict causal attribution, the objective is to isolate the relative contribution of policy-driven price components by separating market prices from state-induced costs.

2 Methodological Framework

To ensure reproducibility and transparency, the analysis is implemented through a structured data-processing pipeline integrating multiple Eurostat datasets covering the period from 1995 to 2024.

2.1 Core Assumptions and Modeling

Since individual household circumstances vary significantly, a standardized "reference household" is defined to maintain comparability across the observation period.

- **Household Composition:** Three-person household (two adults and one child) [14]
- **Living Space:** 70 m^2 residential floor area [4]
- **Heating Demand:** 120 kWh/m^2 annually [1]
- **Electricity Consumption:** 4,000 kWh per household per year [2]

These parameters are consistent with benchmark statistics from the German Federal Environment Agency and Eurostat household consumption surveys.

2.2 Data Sources

The empirical dataset combines the following statistical series:

- Household energy consumption by source [15]
- Electricity prices for household consumers [7],[6]
- Natural gas prices for household consumers [9],[8]
- Equivalised disposable household income distribution [11]

Some of the data are given in semi-annual time intervals. To establish an easier comparison between semi-annual and annual datasets, the semi-annual observations are aggregated to annual averages.

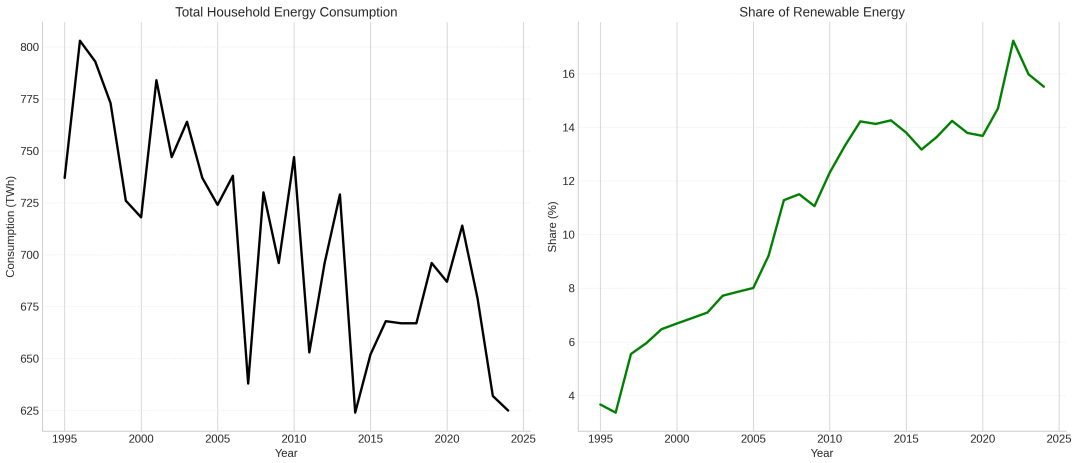

Figure 1: Total household energy consumption in Germany (1995–2024) and reneweable share of household energy consumption in Germany (1995-2024)

3 Structural Development of Household Energy Consumption

Before evaluating socio-economic effects, it is necessary to establish the structural transformation of the residential energy system, which established the baseline for this analysis.
Figure 1 (left) illustrates the development of total household energy demand. Over the examined period, residential energy consumption declined from roughly 750 TWh in the mid-1990s to approximately 650 TWh by 2024, indicating substantial efficiency improvements.
At the same time, the share of renewable energy sources increased considerably (Figure 1 (right)). These two observations indicate a dual success: first, that the overall energy required for daily living is decreasing, and second, that the structural transformation regarding energy sources is effectively taking place. This is an encouraging signal for the energy transition.
However, a critical distinction must be made regarding household energy consumption, which is largely inelastic. While we do observe shifts in consumer behavior, these changes are relatively marginal in the short term. Significant reductions only become visible over a 20-year horizon, suggesting that long-term policies and structural improvements, such as enhanced building insulation and more efficient heating systems, are the primary drivers of change.
The individual resident, particularly one living in a rented apartment, often finds themselves in a position of limited agency. While they can adjust their habits, they cannot unilaterally upgrade the building's infrastructure.

4 Income Distribution Modeling

To evaluate energy poverty across the entire population, a continuous representation of the income distribution is required. The analysis, therefore, fits a lognormal cumulative distribution function to Eurostat income percentile data for each year. The fitted distribution allows estimation of the proportion of households falling below any given income threshold. Household income is then scaled using the modified OECD equivalence scale [10]:

- First adult: 1.0

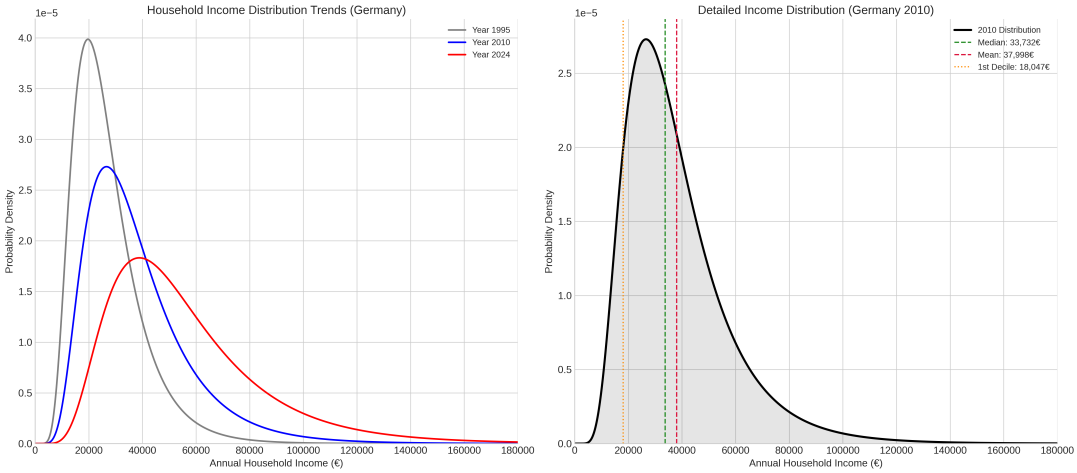

Figure 2: Simulated income distribution for households in Germany

- Second adult: 0.5
- Child: 0.3

This produces an equivalence factor of 1.8 for the modeled three-person household. In Figure 2 you can see examples of the lognormal distribution for the years 1995, 2010 and 2024 as well as the curve for 2010 with more detailes description. The lognormal distribution is defined by two parameters: μ = location parameter and σ = shape parameter.

5 Household Energy Cost Calculation

Energy price data are converted into annual household expenditures by multiplying price per kWh with standardized consumption assumptions. Crucially, the dataset distinguishes between:

- Net energy prices (excluding taxes and levies)
- Gross consumer prices (including taxes and policy-related charges)

Figure 3 shows the widening gap between market prices and final consumer prices, especially for electricity.

6 Quantifying Energy Poverty

Energy poverty is defined as a condition in which a household must spend more than 10% of its disposable income on energy services. For each year, a critical income threshold is calculated:

$$Income_{critical} = 10 \times EnergyCost$$

This threshold is then evaluated within the lognormal income distribution. The resulting cumulative probability corresponds to the share of households whose income falls below this level [12].

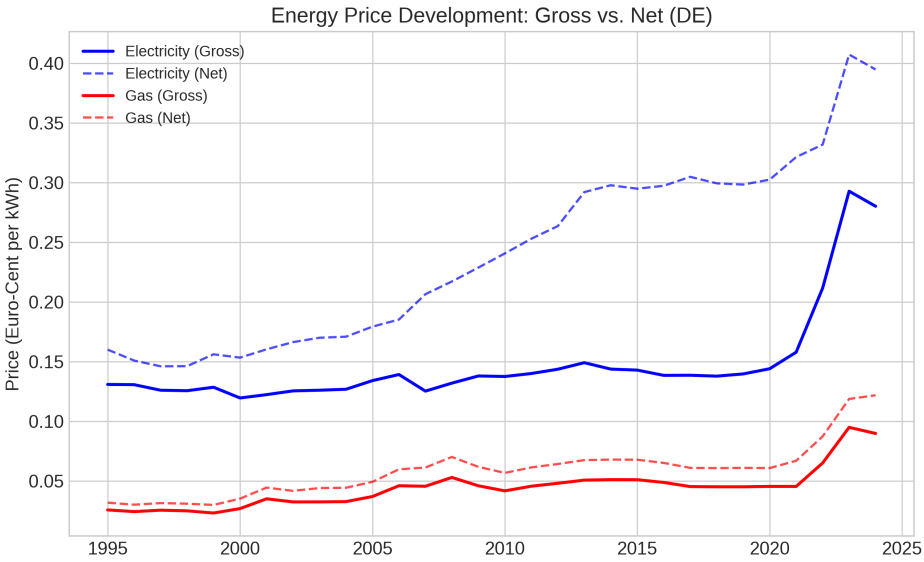

Figure 3: Household energy expenditure with and without fiscal price components

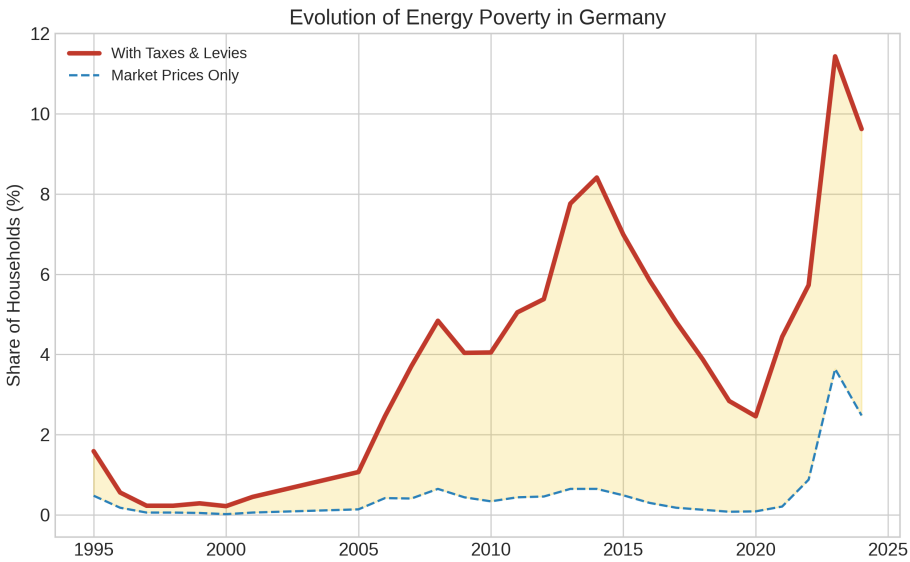

Figure 4: Estimated energy poverty rate in Germany (1995–2024)

Figure 4 illustrates the simulated evolution of energy poverty across the observation period. The data reveals a strong increase: the share of affected households rose from approximately 2% in 1995 to nearly 12% by 2024/2025. This translates to roughly 8–10 million people in Germany living in energy poverty today. Notably, without state-induced taxes and levies, this figure would be only a third as high. While gas prices and associated taxes remained relatively stable until the onset of the war in Ukraine, the primary driver of this development has been the widening disparity between gross and net electricity prices.

6.1 Interpretation and Socio-Economic Implications

The empirical findings illustrated above require careful interpretation regarding their real-world socio-economic impact. Due to the fundamental necessity of electricity and heating for daily survival—ranging from food preservation and digital participation to basic physical health during winter—the price elasticity of demand for residential energy is near zero. Consequently, rising energy costs cannot be offset by simply reducing consumption. These unavoidable costs must be absorbed, directly diminishing the disposable income available for education, healthy nutrition, and social participation, thereby acting as a catalyst for existing social inequalities.

Furthermore, it must be assumed that the actual scope of the problem is notably higher than statistical estimates suggest, indicating a significant unrecorded figure. Households in the lowest income deciles predominantly reside in rented apartments with inherently poor energetic standards. Because tenants lack the authority to upgrade building insulation or replace outdated heating systems, they are caught in a structural investment trap (a split-incentive dilemma). While wealthier homeowners can mitigate rising per-unit costs through capital investments in energy efficiency, such as solar panels or modern heat pumps, low-income renters are structurally locked into paying the escalating prices, forced to bear the full financial brunt of the physical infrastructure transformation required by the energy transition.

These two observations clearly indicate that the structural change towards a climate-friendly energy system disproportionately disadvantages low-income demographics. The empirical data reveals a profound fiscal imbalance: approximately two-thirds of the households currently classified as energy-poor fall into this category solely due to state-induced taxes, levies, and grid expansion fees. Without these policy-driven price components, the energy poverty rate would be reduced by roughly 66 percent. Consequently, while the ecological objectives of the "Energiewende" are progressing, its current financing structure acts as a highly regressive mechanism, shifting the financial weight of the green transition onto those who possess the least economic capacity to absorb it.

7 Policy Evaluation

To mitigate the regressive socio-economic effects of rising energy costs, two distinct policy interventions can be evaluated.

7.1 Exemption of Levies for a Basic Energy Quota

The first approach involves eliminating surcharges and levies for a defined baseline quota of energy consumption. This proposal addresses the fact that energy taxes exert a highly regressive effect, forcing low-income households to spend a disproportionately larger share of their earnings on energy levies compared to wealthier households. Waiving these costs for a basic energy contingent would noticeably reduce the immediate financial burden.

However, this broad-brush approach suffers from significant structural flaws. Primarily, it lacks targeting accuracy: applying a universal tax exemption would lead to substantial fiscal deficits for the state without exclusively benefiting low-income households. More critically, artificially lowering prices for a baseline consumption undermines the ecological steering effect of the energy transition. Lowering the cost of the first 2,000 kWh, for instance, reduces the financial incentive to invest in energy-saving appliances—a phenomenon known in economics as the rebound effect. Balancing social relief with the necessary ecological price signals to meet climate targets remains the core challenge of this model.

7.2 Targeted Fiscal Refund Mechanism

A more economically sound approach involves a targeted refund for households whose energy expenditures exceed the 10% threshold. Under this model, households receive a direct transfer corresponding exactly to the amount required to push their energy burden back to the 10% limit:

$$Subsidy = EnergyCost - (0.10 \times HouseholdIncome)$$

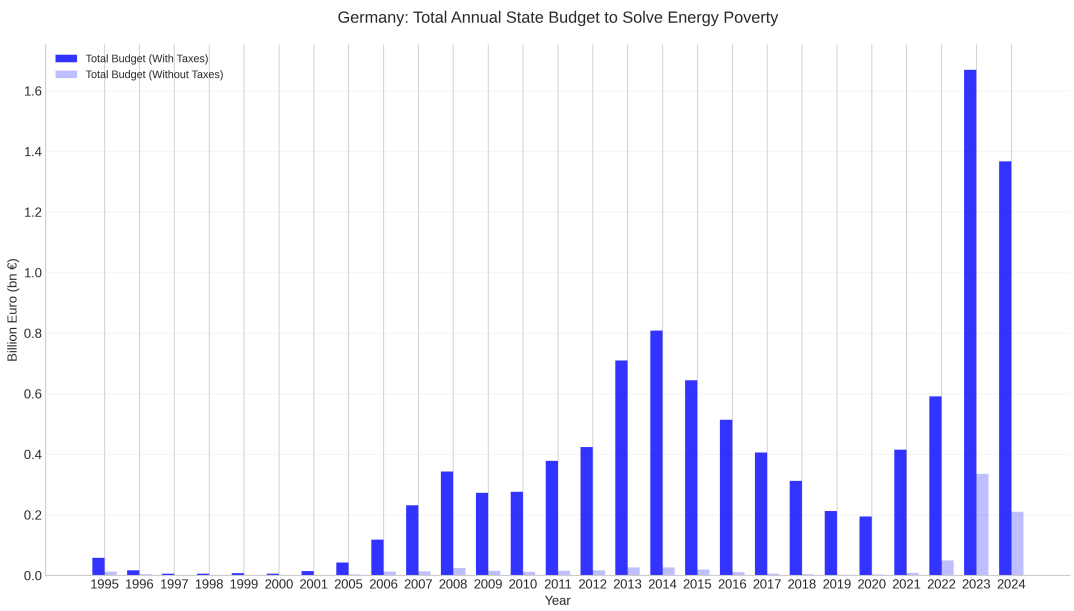

Figure 5: Simulation of targeted fiscal refund mechanism

The primary advantage of this second proposal is that the market price mechanism remains entirely intact. Consequently, macroeconomic efficiency is preserved, and no deadweight welfare

loss occurs. Furthermore, this mechanism is highly precise. The refund could be administered with minimal bureaucracy, for example, directly through the annual income tax return system. Because state funds are deployed exclusively where the financial burden threshold is actually breached, this model is far more cost-efficient than a universal subsidy.
Based on the simulated income distribution and assuming approximately 30 million households in Germany, scaling this targeted refund mechanism to completely absorb energy poverty would cost the state roughly €1.6 billion for the year 2024. Compared to the budgets of other social policy measures, this represents a highly targeted, cost-efficient, and financially viable instrument to cushion the social hardships of the transition.

8 Conclusion

The empirical analysis suggests that the German energy transition has produced measurable distributional effects. Rising energy prices, particularly those driven by fiscal components, have increased the share of households facing excessive energy expenditure burdens. While climate policy remains essential for long-term sustainability, the findings highlight the importance of distributionally balanced policy design. Targeted fiscal compensation mechanisms offer a promising approach to maintaining both climate policy effectiveness and social equity, without attacking the free market and leaving price-finding mechanisms fully intact.

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